2023-2024 春季 线代期中 仅供参考

一、D D C C A

二、(1) $A^{-1}=\begin{bmatrix} 1 & & \\ -a & 1 & \\ (3a-b)/2 & -3/2 & 1/2 \end{bmatrix}$

(2) 2

(3) $4^{2023}A=4^{2023}\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 2 & -2 & 2 \end{bmatrix}$

(4) $\begin{bmatrix} \frac{7}{3} \\ \\ \frac{10}{3} \end{bmatrix}$

(1) $[A\ 1]=\left[\begin{array}{ccc|ccc} 1 & & & 1 & & \\ a & 1 & & & 1 & \\ b & 3 & 2 & & & 1 \end{array}\right]\to\left[\begin{array}{ccc|ccc} 1 & & & 1 & & \\ 0 & 1 & & -a & 1 & \\ 0 & 3 & 2 & -b & 0 & 1 \end{array}\right]\to\left[\begin{array}{ccc|ccc} 1 & & & 1 & & \\ 0 & 1 & & -a & 1 & \\ 0 & 0 & 2 & 3a-b & -3 & 1 \end{array}\right]$

$\to\left[\begin{array}{ccc|ccc} 1 & & & 1 & & \\ & 1 & & -a & 1 & \\ & & 1 & (3a-b)/2 & -3/2 & 1/2 \end{array}\right]=[1\ \ A^{-1}]$

(2) 法1 🐼 令 $A_{4\times3}=\begin{bmatrix} 1 & & \\ & & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $AB=\begin{bmatrix} 1 & & \\ & & 1 \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\Rightarrow r(AB)=2$

法2：$R(A)=2$，$R(B)=3$

$R(AB)\geq R(A)+R(B)-n=2+3-3=2$

$R(AB)\leq \min\{R(A),R(B)\}=2$

$\Rightarrow R(AB)=2$

(3) 算！A^2，A^3，即得出规律

(4) $A=\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ $b=\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$

三、 $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 1 \\ 1 & -4 & -7 \end{bmatrix}$

$$E_{21}(-2)A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & -5 \\ 1 & -4 & -7 \end{bmatrix}$$

$$E_{31}(-1)E_{21}(-2)A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & -5 \\ 0 & -2 & -10 \end{bmatrix}$$

$$E_{32}(-2)E_{31}(-1)E_{21}(-2)A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = E_{21}^{-1}(-2)E_{31}(-1)^{-1}E_{32}^{-1}(-2)\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 1 & 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 0 \end{bmatrix} = LU$$

四.(a)

① $C(A)$，对A行变换

$$A \to \begin{bmatrix} 0 & \textcircled{2} & 0 & 3 & 0 \\ 0 & 0 & \textcircled{2} & -1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{10} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow C(A) = \text{span}\left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 10 \\ -5 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \\ 2 \\ 7 \end{bmatrix} \right\}$$

② $C(A^T)$ 对A^T行变换

$$A^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 2 & 1 & 4 & -1 \\ 4 & 1 & 10 & -5 \\ 1 & 1 & 1 & 1 \\ 6 & 3 & 2 & 7 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 3 & -6 & 9 \\ 0 & 3 & 4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} \textcircled{1} & 1 & 1 & 1 \\ & \textcircled{1} & -2 & 3 \\ & & \textcircled{10} & -10 \\ & & & 0 \\ & & & 0 \end{bmatrix} \Rightarrow C(A^T) = \text{span}\left\{ \begin{bmatrix} 0 \\ 2 \\ 4 \\ 1 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 10 \\ 1 \\ 2 \end{bmatrix} \right\}$$

③ $N(A)$

$$Ax = 0 \Rightarrow \begin{bmatrix} 0 & \textcircled{2} & 0 & 3 & 0 \\ 0 & 0 & \textcircled{2} & -1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{10} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} = 0 \Rightarrow \begin{cases} x_1, x_4 \in \mathbb{R} \\ x_2 = -\frac{3}{2}x_4 \\ x_3 = \frac{1}{2}x_4 \\ x_5 = 0 \end{cases} \Rightarrow x = k_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + k_4 \begin{bmatrix} 0 \\ -3/2 \\ 1/2 \\ 1 \\ 0 \end{bmatrix} \quad (k_1, k_2 \in \mathbb{R})$$

④ $N(A^T)$

$$A^T y = 0 \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ & 1 & -2 & 3 \\ & & 10 & -10 \\ & & & 0 \\ & & & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = 0 \Rightarrow \begin{cases} y_1 = -y_4 \\ y_2 = -y_4 \\ y_3 = y_4 \\ y_4 \in \mathbb{R} \end{cases} \Rightarrow y = k_0 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix} \quad k_0 \in \mathbb{R}$$

(b) $Ax = b$ 时 利用高斯消元化简增广矩阵(略)

令出一特解 $x_p = [1 \quad -5 \quad 1 \quad 3 \quad 1]^T$

$$x = x_p + x_n = \begin{bmatrix} 1 \\ -5 \\ 1 \\ 3 \\ 1 \end{bmatrix} + k_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + k_4 \begin{bmatrix} 0 \\ -3/2 \\ 1/2 \\ 1 \\ 0 \end{bmatrix} \quad k_1, k_4 \in \mathbb{R}$$

五、令 $X=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$，

则 $XA=\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}=\begin{bmatrix} a & a+2b \\ c & c+2d \end{bmatrix}$

$AX=\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}=\begin{bmatrix} a+c & b+d \\ 2c & 2d \end{bmatrix}$

$T(x)=XA+AX=\begin{bmatrix} 2a+c & a+3b+d \\ 3c & c+4d \end{bmatrix}=(2a+c)v_1+(a+3b+d)v_2+(3c)v_3+(c+4d)v_4$

$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)=(v_1\ v_2\ v_3\ v_4)\begin{pmatrix} 2a+c \\ a+3b+d \\ 3c \\ c+4d \end{pmatrix}$

六、证明：$(A+B)^2$

$=(A+B)(A+B)=A^2+AB+BA+B^2=A+B$

$\because A^2=A,\ B^2=B$

$\therefore AB+BA=0$ ①

$B(AB+BA)=BAB+B^2A=(BA)B+B^2A=-AB^2+B^2A=0$

又 $\because B^2=B$

$\therefore -AB^2+B^2A=-AB+BA=0$ ②

联立①② $\begin{cases} AB+BA=0 \\ -AB+BA=0 \end{cases}\Rightarrow AB=0$，得证

七、(a) $(AB)^2=(AB)(AB)=\begin{bmatrix} 8 & 0 & -4 \\ -\frac{3}{2} & 9 & -6 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 8 & 0 & -4 \\ -\frac{3}{2} & 9 & -6 \\ -2 & 0 & 1 \end{bmatrix}=\begin{bmatrix} 72 & 0 & -36 \\ -\frac{27}{2} & 81 & -54 \\ -18 & 0 & 9 \end{bmatrix}$

(b) $(AB)^2=9(AB)$

$R(A)\geq R(AB)=2\Rightarrow R(A)=2$，同理 $R(B)=2$

A行满秩，令 $XA=I_2$　X为A左逆

B列满秩，令 $BY=I_2$　Y为B右逆

$\therefore BA=(XA)(BA)(BY)=X(AB)^2Y=9XABY=9I$

地精爱你！